INCH-POUND

MIL-P-70674 (AR)
AMENDMENT 1
3 April 1989

MILITARY SPECIFICATION

PROJECTILE 155MM: XM864 METAL PARTS
FOR BASE BURNER ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-70674 (AR), dated 4 April 1988 and is approved for use within the US Army Armament Munitions and Chemical Comand and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Add the designation "(M864)" in the title and change "XM864" to "M864" wherever it appears in the specification.

PAGE 2

2.2 Other publications.

Add the following:

"AMERICAN NATIONAL STANDARD INSTITUTE

ANSI Z1.1 - Guide for Quality Control Charts
ANSI Z1.2 - Control Chart Method of Analyzing Data
ANSI Z1.3 - Control Chart Method of Controlling Quality During Production

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018)"

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4.4.2 Add the following new paragraphs:

"4.4.2.c Statistical process control (SPC). For those dimensions identified in the Classification of Defects Tables by referencing Appendix A, the contractor must utilize the principals of SPC as defined herein, ANSI Z1.1, ANSI Z1.2, ANSI Z1.3 and Appendix A. When the contractor satisfies the control conditions identified in Appendix A acceptance will then be by SPC. The SPC sample size and frequency of selection shall be determined by the contractor. If the SPC sample is to be less than 4 items per hour, then the approval of the procuring agency must be sought prior to the elimination of sampling inspection for acceptance. The contractor may request to implement SPC acceptance for any other feature that is not

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annotated in the classification for defects by requesting approval of the procurement agency and by meeting all the conditions specified in this specification. When not accepting product by SPC, acceptance of product for those features identified by referencing Appendix A only be in accordance with MIL-STD-414 or MIL-STD-1235 plans CSP-1, -2, -T using variable measurement capable gages."

PAGE 8

4.4.2.1 For each of the following defect characteristics delete "min" and append an "Appendix A" in the requirement paragraph column 101, 102.

PAGE 9

4.4.2.2 For each of the following defect characteristics append an "Appendix A" in the requirement paragraph column for items 101, 103, 104, 105, 106, 107 and 108.

PAGE 10

4.4.2.3 For each of the following defect characteristics append an "Appendix A" in the requirement paragraph column for items 101, 102, 104 and 106.

PAGE 11

4.4.2.4 For each of the following defect characteristics append an "Appendix A" in the requirement paragraph column for items 102, 104, 106, 110, 114 and 115.

PAGE 13

5.2 Delete second and third sentences and substitute the following new sentences:

"In addition, packaging shall be, as a minimum, single item and moistureproof using heat sealed barrier bags in accordance with the Level B requirements of MIL-STD-1169. Additional packaging shall be provided as required to prevent wearing of phosphate coating during shipment."

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Attach the attached Appendix A to the back of the specification.

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APPENDIX A

10. SCOPE

10.1 Scope. This appendix details the procedure for implementation of statistical process control for articles covered in this specification. This Appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Application of statistical process contol. The contractor will, for the identified dimensions, maintain Xbar and R SPC charts. After the contractor has demonstrated that his process for manufacturing a particular feature is in control, as defined in 30.1.1; he will discontinue acceptance sampling inspection in favor of acceptance by SPC. However, SPC charting will be maintained for all identified dimensions whether or not the process is in control. It is expected that the contractor will make every effort to bring each process into control. It is required that the contractor re-evaluate his established control limits on a monthly basis and make those changes warranted by the data. At no time should his process parameters vary such that it will fail to meet the requirements established in subsequent paragraphs. Should this happen, the contractor will return to acceptance by sampling.

30.1.1 Implementation of SPC for acceptance of product. For those dimensions identified in the classifications of defects tables as being applicable for elimination of lot sampling acceptance, the contractor shall implement, on a dimension by dimension basis, acceptance by SPC after he conforms to each of the following steps:

a. The contractor performs an in depth process capability study to establish the mean process average (Xdouble bar), Rbar, upper and lower control limits. This study will consist of at least 40 subgroups of data collection.

b. The contractor demonstrates that his process is under control for at least 20 consecutive production days. Control is defined as remaining within the established control parameters and not exhibiting any of the out of control conditions defined in 30.1.2.a thru 30.1.2.e more than two times on an individual character basis.

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APPENDIX A

c. Thirty days consecutive production have passed acceptance sampling for the feature.

d. The process has a demonstrated Cpk equal to or greater than 1.3 as demonstrated when conducting the above test. Cpk is defined as follows:

$$Cpk = \frac{\text{process mean - nearest specification limit}}{3 \text{ sigma}}$$

sigma = process standard deviation (calculated by the range method per the referenced ANSI documents)

e. The contractor documents his detailed process description. This process description must be submitted to Commander, AMCCOM, Attn: AMSMC-QAR-Q (D), Picatinny Arsenal, New Jersey 07806-5000 for review and disapproval if the description does not meet its objectives. That is that the process description will identify all known contributing variables and their respective controls that are necessary to maintain the specific dimension (i.e., weight, temperature, machine speed, time, tool life, solution proportion, etc.). The government will furnish the contractor with written notice of the acceptability of his process controls. A notice of control acceptability is required prior to the implementation of acceptance by SPC.

30.1.2 Process control analysis. Once SPC acceptance is implemented, the contractor will evaluate his process for evidence of non-random variation (e.g. trends, extreme values, etc.). Should any out of control condition be detected, the product produced from the last sample until the process is brought back into control will be isolated for analysis. The contractor will bring the process back into control or shut the process down. Upon determining the cause of the out of control condition, the contractor will determine if the product manufactured and isolated could be out of engineering tolerance. If the out of control condition did not result in the production of nonconforming material, the material will be released. If the out of control condition could have created nonconforming product, the contractor will sample the product using MIL-STD-105 and the AQL specified in the classification of defects table to determine the acceptability of the isolated product. This evaluation of out of control product requires the review and approval of the local government quality assurance representative. As a minimum, the contractor will look for any of the following out of control indications:

a. If any point goes outside the Xbar or R chart control limits.

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b. If 9 points in a row are either continually above or continually below the X double bar.

c. If 6 points in a row are either continually increasing or continually decreasing on the Xbar chart.

d. If 6 points in a row are increasing on the range chart.

e. If any individual sample, Xi, falls outside of the specification limits.

30.1.2.1 Process capability review. At least once a month the contractor will review his process data and establish updated control limits.

30.1.3 Mandatory return to sampling acceptance. Acceptance by SPC will be discontinued in accordance with the following paragraphs. In any case, control charting will be continued. When this occurs, the contractor may reintroduce acceptance by SPC by repeating the steps outlined in paragraph 30.1.1.

30.1.3.1 Out of control incidents. If for any individual process, an out of control incident is identified 3 times within 5 consecutive shifts or 5 times within 30 consecutive shifts; and the analysis indicated that nonconforming product may have been manufactured, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.3.2 Process interruption. If the process is interrupted for more than 10 calendar days, the contractor will return to acceptance by sampling for that pertinent feature.

30.1.3.3 Loss of control. If either by testing or data review it is determined that the Cpk is no longer at the required minimum of 1.3, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.4 Records. The contractor will maintain all records dealing with SPC in his files for a minimum of 5 years after completion of the contract.

30.2 Government auditing. In order to verify the state of control of the process, it is recommended that government representatives select samples at unscheduled times. The sample mean and range should then be compared to the data being recorded by the contractor. If the data is not consistent with the reported data, corrective action should be sought and, if necessary, screening of material should be required.

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